

Math 105LA Assignment 3

Fall 2019

Due Sun 10/20, 11pm

Write a MATLAB function to implement the Muller's method.

Input: f , p_0 , p_1 , p_2 , N , tol , tol_fp

Output: p

Step 1: Set $n = 1$, $q_0 = f(p_0)$, $q_1 = f(p_1)$, $q_2 = f(p_2)$

Step 2: While $n \leq N$, repeat step 3 to step 8

Step 3: Set $a = ((p_0 - p_2) * (q_1 - q_2) - (p_1 - p_2) * (q_0 - q_2)) / ((p_1 - p_2) * (p_0 - p_2) * (p_1 - p_0))$
 $b = ((p_1 - p_2)^2 * (q_0 - q_2) - (p_0 - p_2)^2 * (q_1 - q_2)) / ((p_1 - p_2) * (p_0 - p_2) * (p_1 - p_0))$
 $c = q_2$
 $D = \text{sqrt}(b^2 - 4 * a * c)$

Step 4: if $|b+D| > |b-D|$
 Set $p = p_2 - 2 * c / (b+D)$
 else
 Set $p = p_2 - 2 * c / (b-D)$
 end

Step 5: Set $q_0 = q_1$, $q_1 = q_2$, $q_2 = f(p)$

Step 6: If $|q_2| < \text{tol_fp}$
 Output('f(p) is smaller than tol_fp', p), stop
 elseif $|p - p_2| < \text{tol}$
 Output('p is converged with $|p_{n+1} - p_n|$ smaller than tol', p) stop
 end

Step 7: Set $n = n + 1$

Step 8: Set $p_0 = p_1$, $p_1 = p_2$, $p_2 = p$;

Step 9: Output('Maximum number of iteration is reached' ,p)

You can start with the template below. Make sure that you use the **same order** of input:

```
function p = muller(f,p0,p1,p2,N,tol,tol_fp)
% f = the function to be solved
% p0,p1,p2 = initial guesses
% tol = tolerance
% N = maximum number of iterations
end
```

The following is some example for testing the code.

Input	Output
$f = @(x) x^2 - 2 * x - 400$; $p_0 = 1$; $p_1 = 2$; $p_2 = 3$; $p = \text{muller}(f, p_0, p_1, p_2, 1, 1e-10, 1e-10)$	$ f(p) $ is smaller than $1.00e-10$ $p = 21.0250$
$f = @(x) x^4 - x^3 + x^2 + x + 1$; $p_0 = 0.5$; $p_1 = -0.5$; $p_2 = 0$; $p = \text{muller}(f, p_0, p_1, p_2, 300, 1e-5, 1e-5)$ $\text{fp} = \text{abs}(f(p))$ $\text{fp_star} = \text{abs}(f(\text{conj}(p)))$	$ f(p) $ is smaller than $1.00e-05$ $p = -0.4334 - 0.5258i$ $\text{fp} = 9.4756e-06$ $\text{fp_star} = 9.4756e-06$
$f = @(x) -\exp(x) * 1.5 + x^2 - x + 10$; $p_0 = 0.5$; $p_1 = -0.5$; $p_2 = 0$; $p = \text{muller}(f, p_0, p_1, p_2, 1, 1e-5, 1e-5)$ $p = \text{muller}(f, p_0, p_1, p_2, 10, 1e-5, 1e-5)$ $\text{abs}(p - \text{conj}(p))$	Maximum number of iteration is reached. $p = 5.4714 - 2.5201i$ $ f(p) $ is smaller than $1.00e-05$ $p = 2.1067 - 0.0000i$ $\text{ans} = 5.4042e-08$

Submit a single file **named by your UCInetID** (e.g. wtleung.m) that contains the function. Your code will be tested on a few problems. Your grade of this assignment is calculated by the number of problems that your code solves correctly. Make sure your function runs without errors and has **the same order of input** above. **Failure to do so will result in a zero grade.**